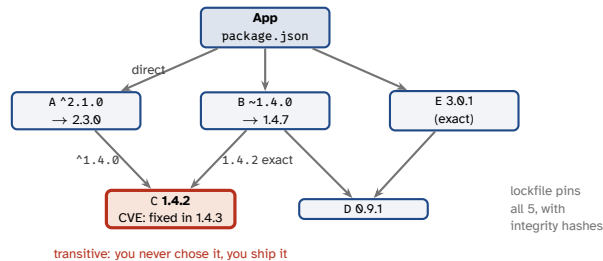


**In one line:** Every dependency is code you ship, cannot fully read, and must keep updating. **Semver ranges** say what you *accept*; the **lockfile** records what you *got* (every transitive package, exact version + hash) so every machine and CI builds the same tree. Choose libraries by **maintenance, size, licence and exit cost**, update in **small automated steps**, and remember that most of your tree is **transitive** — you are responsible for packages you never chose.

#### A vulnerable transitive dependency — and the fix ladder



1 find the path: npm explain C / npm ls C - swift package show-dependencies

not exploitable in your usage? record why, with an expiry — don't just silence the scanner

2 update the direct dep: a B release that allows C ≥ 1.4.3 — best: upstream fixed

3 override the transitive: npm overrides, Yarn resolutions, pnpm overrides; test, add a removal note

SPM: one version of C per graph — incompatible ranges fail resolution. npm: may nest two copies side by side

4 patch / fork (patch-package, a pinned fork) — you now own it

5 remove the feature or the dependency

### Package managers

| Tool      | Lockfile          | Model                                                                                                         |
|-----------|-------------------|---------------------------------------------------------------------------------------------------------------|
| SPM       | Package.resolved  | built into Xcode/swift; source + binaryTarget (XCFramework + checksum); one version per package               |
| CocoaPods | Podfile.lock      | Ruby; generates a workspace; trunk read-only from 2 Dec 2026 (existing pods still install) — RN still uses it |
| Carthage  | Cartfile.resolved | builds frameworks, you link them; decentralised, rarely chosen now                                            |
| npm       | package-lock.json | nested node_modules: two versions can coexist                                                                 |
| Yarn      | yarn.lock         | Berry adds Plug'n'Play; RN projects use nodeLinker: node-modules                                              |
| pnpm      | pnpm-lock.yaml    | content-addressed store + symlinks; <b>strict</b> : undeclared ("phantom") imports fail                       |

### Semver and ranges

- MAJOR.MINOR.PATCH = breaking . feature . fix; 0.y.z: anything may change. A promise by humans, not a guarantee — tests decide.
- npm: ^1.2.3 => 1.2.3 <2.0.0 . ^0.2.3 => 0.2.3 <0.3.0 (caret treats the first non-zero as major) . ~1.2.3 => 1.2.3 <1.3.0 . 1.2.3 exact. Pre-releases (2.0.0-beta.1) are skipped by ranges unless asked for.
- SPM: from: "1.2.3" = 1.2.3...<2.0.0 . upToNextMinor(from:) = 1.2.3...<1.3.0 . exact: . branch/revision (not for releases).
- Apps:** pin native-heavy deps exactly (RN, SDKs) and let the lockfile pin the rest. **Libraries:** wide ranges, peers for shared singletons (react, react-native).

### Lockfile rules

Commit it (apps); CI installs *from* it (npm ci, pod install, resolved SPM) and fails if it drifts. Review lockfile diffs: a one-line change in package . json can move 200 transitive versions. A library's lockfile is not published — consumers resolve its ranges themselves.

### Example — ranges, override, SPM

```
// package.json (comments for the sheet only)
"dependencies": {
  "react-native": "0.81.0", // exact: native, upgrade on purpose
  "date-fns": "^4.1.0", // >=4.1.0 <5.0.0
  "@example/analytics": "~2.3.1" // >=2.3.1 <2.4.0
},
"overrides": {
  "semver": "7.5.4" // force a patched transitive
}
// Package.swift
.package(url: "https://example.com/a.git", from: "1.1.0")
// 1.1.0..<2.0.0
.package(url: "https://example.com/b.git",
        .upToNextMinor(from: "0.9.2")) // 0.9.2..<0.10.0
```

### Remember

**"Ranges say what you accept, the lockfile what you got; every dependency is yours to update — or remove."**

### Choosing a library — the signals

- Maintenance:** releases in the last months, issues and PRs answered, CI green, a changelog, honest semver; supports the current OS, Swift 6 / RN New Architecture, ships types.
- Bus factor:** more than one active maintainer, or a company behind it; who fixes a CVE at 2 am?
- Size + cost:** binary size delta (App Thinning Size Report, JS bundle analyser), transitive count, launch cost (dynamic frameworks), permissions and privacy manifest it brings.
- Need it?** 20 lines of your own beat a package for a helper; platform APIs (URLSession async/await, fetch, Intl) replace many classics. **Exit cost:** wrap it behind your interface.

### Licences — what they cost an app

|            |                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------|
| MIT, BSD   | permissive: keep copyright + licence text (Acknowledgements screen)                                          |
| Apache-2.0 | permissive + <b>patent grant</b> ; keep NOTICE; not GPLv2-compatible                                         |
| MPL-2.0    | file-level copyleft: publish changes to <i>its</i> files only                                                |
| LGPL       | users must be able to relink a modified library — hard with static linking in a signed iOS app; legal review |
| GPL / AGPL | the combined app must ship as GPL source (AGPL: network use too) — a no for closed apps                      |
| no licence | all rights reserved: you may not use it                                                                      |

Gate it in CI: a licence scanner emits SPDX ids, checked against an allow-list.

### Vendor, update, remove

- Vendor** (copy source/binary into the repo) for tiny, abandoned or patched code, or offline builds — you now own its updates and CVEs. Middle ground: pinned fork, patch-package.
- Update:** Renovate / Dependabot PRs on a schedule; group patch + minor, majors one by one with the changelog read; automerge only low-risk groups on green CI. RN/Xcode upgrades are planned work.
- Remove:** find imports, swap in the platform API, delete, check the lockfile and binary shrank; knip / depcheck find unused JS deps.

### Traps

- Two copies of react in an RN monorepo → "Invalid hook call": dedupe; libraries declare it as a **peer**.
- "We pinned ^1.2.3" — a caret range is not a pin; the lockfile is.
- Code relying on a transitive package it never declared — breaks under pnpm or the next update.